

Variational Quantum Bayesian Regression

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Abstract—
Index Terms—

characterizes accuracy and optimization stability as a function of shot budget, ansatz depth, and optimizer choice.

I. INTRODUCTION

Bayesian linear regression (BLR) provides a principled way to incorporate prior knowledge and to quantify predictive uncertainty, with broad use in signal processing, system identification, and uncertainty-aware learning [1], [2]. In its standard Gaussian form, BLR reduces to a strongly convex quadratic MAP estimation problem. We formalize this setting in Problem 1, where the objective combines a data-fit term and a prior-precision quadratic.

Problem 1 (Bayesian linear regression training). *Given training data $(\mathbf{X}, \mathbf{y}) \in \mathbb{R}^{N \times D} \times \mathbb{R}^N$ and a Gaussian prior $(\mathbf{m}_0, \Sigma_0) \in \mathbb{R}^D \times \mathbb{R}^{D \times D}$, and error noise distribution $\epsilon \sim \mathcal{N}(0, V)$, we aim to find the optimal weights $\mathbf{w}^*$ such that*

$$\min_{\mathbf{w} \in \mathbb{R}^D} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + (\mathbf{w} - \mathbf{m}_0)^\top V \Sigma_0^{-1} (\mathbf{w} - \mathbf{m}_0). \quad (1)$$

The closed-form solution of (1) is given as

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + V \Sigma_0^{-1})^{-1} (\mathbf{X}^\top \mathbf{y} + V \Sigma_0^{-1} \mathbf{m}_0), \quad (2)$$

which has the classical running time of $\mathcal{O}(ND + ^2 + D^3)$ for forming the gram matrix $\mathbf{A} = (\mathbf{X}^\top \mathbf{X} + V \Sigma_0^{-1})$, the vector $\mathbf{b} = \mathbf{X}^\top \mathbf{y} + V \Sigma_0^{-1} \mathbf{m}_0$ and followed by classical inversion such as singular value decomposition (SVD). Also, computing the gram matrix requires computing the inversion of Σ_0 beforehand which also cost $\mathcal{O}(D^{-3})$ under classical matrix inversion.

Contributions. In this paper, we explore a *variational training paradigm* for Problem 1 inspired by recent progress in NISQ variational linear-algebra primitives. Rather than constructing the normal equations, we build a measurement-driven estimator of the MAP objective $\mathcal{L}(\mathbf{w})$ using row-wise access to data states and overlap estimation for the data-fit terms, together with computational-basis sampling for the diagonal-prior quadratic term, and then train a parameterized circuit so that its output state encodes the *direction* of the MAP solution in amplitude form. In particular, we propose *Variational Quantum Bayesian Regression (VQBR)*, a hybrid algorithm that outputs a circuit preparing a state whose amplitudes approximate $\mathbf{w}^*/\|\mathbf{w}^*\|_2$, introduce a direct MAP-objective estimator that avoids explicit Gram-matrix construction, and outline an NISQ-oriented evaluation protocol that

II. RELATED WORK

A. Classical BLR, MAP, and computational regimes

Bayesian linear regression (BLR) with Gaussian priors yields a strongly convex quadratic MAP problem with a closed-form optimizer and posterior covariance [1], [2]. In large feature dimension D , the computational bottleneck is often not the formula itself but the cost of forming and factoring normal-equation objects (e.g., $\mathbf{X}^\top \mathbf{X}$) or repeatedly applying $\mathbf{X}$ and $\mathbf{X}^\top$ in iterative solvers. This motivates regimes where one seeks (i) matrix-free optimization, (ii) implicit access to the solution through prediction primitives, or (iii) structured priors enabling cheaper quadratic terms.

B. Fault-tolerant quantum least squares and linear regression

Quantum algorithms for least squares and linear regression originate from quantum linear-systems methods (QLSAs) and “prediction by regression” formulations that aim to output inner products or predictions without explicitly outputting all D coordinates [3]–[5]. More recent developments study regularized least squares under block-encoding / QSVT-style access models [6]. At the same time, a growing body of work refines the practical limits of regression speedups, including quantum-inspired algorithms and “dequantization-style” perspectives which reduce or remove exponential advantages under comparable data-access assumptions [7], [8].

C. NISQ variational linear-algebra primitives

Near-term (NISQ) constraints have motivated hybrid variational solvers that train parameterized circuits using measurement-defined costs. A central example is the variational quantum linear solver (VQLS), which trains $U(\theta)|0\rangle$ so that $A|x(\theta)\rangle$ aligns with $|b\rangle$, with costs estimated via overlap-type circuits [9]. Our work borrows the “overlap-estimation as a training primitive” philosophy, but differs in the optimization target and access pattern: rather than requiring an explicit operator representation (e.g., block-encoding) for $\mathbf{A} = \mathbf{X}^\top \mathbf{X} + \Sigma_0^{-1}$, we estimate the BLR MAP objective *directly* from row-wise state preparation of $\{|x_i\rangle\}$ and a prior state, plus computational-basis sampling for the diagonal prior quadratic term.

D. Quantum models for Bayesian learning and uncertainty

Beyond point-estimation regression, several quantum approaches target Bayesian learning or uncertainty quantification. Quantum circuits have been proposed as variational families for approximate inference in probabilistic models (variational inference with quantum circuits) [10], while other works discuss Bayesian learning perspectives on quantum computation more broadly [11]. In parallel, Bayesian uncertainty in regression is often studied through Gaussian processes (GPs); recent works propose quantum-kernel GP regression variants that aim to retain predictive variance information [12]. These directions are complementary to our focus: VQBR targets the BLR MAP *direction state* under an explicit Gaussian prior, with uncertainty extraction (posterior variance) left as future work.

E. Overlap/amplitude estimation as a measurement primitive

The measurement budget required to estimate overlaps and expectation values is a key limiter in NISQ training loops. While naive sampling yields $O(1/\epsilon^2)$ shots for additive error ϵ , there is active work on reducing depth or improving constant factors via variational amplitude-estimation-style primitives [13]. Our numerical study therefore emphasizes accuracy–cost tradeoffs as a function of shot budget and ansatz depth.

Positioning of VQBR. In summary, VQBR sits at the intersection of (i) BLR/MAP objectives with explicit priors, and (ii) NISQ variational linear-algebra training via overlap measurements. Its distinguishing feature is a *direct MAP-objective estimator* that avoids explicit Gram construction and avoids requiring a full operator encoding for $\mathbf{X}^\top \mathbf{X} + \Sigma_0^{-1}$, while naturally producing a normalized solution-direction state that supports inner-product prediction queries.

III. VARIATIONAL QUANTUM BAYESIAN REGRESSION TRAINING ALGORITHM (VQBR)

A. Problem statement

Problem 2 (Quantum Bayesian linear regression training). *Given training data $(\mathbf{X}, \mathbf{y}) \in \mathbb{R}^{N \times D} \times \mathbb{R}^N$, a Gaussian prior $(\mathbf{m}_0, \Sigma_0) \in \mathbb{R}^D \times \mathbb{R}^{D \times D}$, and observation noise $\epsilon \sim \mathcal{N}(0, V)$, design a parameterized quantum circuit $U(\theta)$ such that, when executed, a designated d -qubit work register prepares a state $|\phi(\theta^*)\rangle$ whose amplitudes encode (up to normalization) the solution of Problem 1, i.e.,*

$$|\phi(\theta^*)\rangle \approx \left| \frac{\mathbf{w}^*}{\|\mathbf{w}^*\|_2} \right\rangle, \quad (3)$$

where $\mathbf{w}^*$ is the minimizer of (1).

Here we assume that prior precision matrix Σ_0^{-1} is given or effectively computed using quantum singular value transformation [14], [15]. Let $\mathbf{c} := V\Sigma_0^{-1}\mathbf{m}_0$, given Σ_0^{-1} , which can be computed classically using matrix-vector multiplication in $\mathcal{O}(D^2)$ or effectively using quantum procedures [16], [17]. We propose VQBR, a hybrid (NISQ-friendly) training procedure that learns parameters θ of a circuit $U(\theta)$ whose output state $|\phi(\theta)\rangle$ encodes the quantum version of $\mathbf{w}^*$. Throughout, let

$d = \lceil \log_2 D \rceil$ and interpret feature vectors as zero-padded to length 2^d when needed.

B. Quantum encoding variational objective

VQBR leverages the fact that quantum states are normalized, whereas the optimizer $\mathbf{w}^*$ is an unconstrained vector. We therefore decouple a *direction* (represented by a normalized quantum state) from a *scalar scale*. Let $|\phi(\theta)\rangle = U(\theta)|0\rangle^{\otimes d}$ and let $\phi(\theta) \in \mathbb{R}^{2^d}$ denote its (real) amplitude vector, so $\|\phi(\theta)\|_2 = 1$. We parameterize candidate weights along the ray spanned by $\phi(\theta)$ as

$$\mathbf{w}(t, \theta) = t\phi(\theta), \quad t \in \mathbb{R}, \quad (4)$$

and define the ray-parameterized MAP objective $\mathcal{L}(t, \theta) := \mathcal{L}(\mathbf{w}(t, \theta))$. Substituting (4) into the objective of equation (1) and dropping constants independent of t and θ yields a 1D quadratic in t :

$$\mathcal{L}(t, \theta) = \alpha(\theta)t^2 - 2\beta(\theta)t + \text{const}, \quad (5)$$

with

$$\begin{aligned} \alpha(\theta) &:= \phi(\theta)^\top \mathbf{X}^\top \mathbf{X} \phi(\theta) + \phi(\theta)^\top V \Sigma_0^{-1} \phi(\theta), \\ \beta(\theta) &:= \phi(\theta)^\top \mathbf{X}^\top \mathbf{y} + \phi(\theta)^\top V \Sigma_0^{-1} \mathbf{m}_0. \end{aligned} \quad (6)$$

Since $V\Sigma_0^{-1} \succ 0$, we have $\alpha(\theta) > 0$ for all θ . Therefore, for any fixed θ , the minimizer over t exists, is unique, and is available in closed form:

$$t^*(\theta) = \arg \min_{t \in \mathbb{R}} \mathcal{L}(t, \theta) = \frac{\beta(\theta)}{\alpha(\theta)}. \quad (7)$$

Eliminating t yields the reduced (direction-only) objective

$$\tilde{\mathcal{L}}(\theta) := \min_{t \in \mathbb{R}} \mathcal{L}(t, \theta) = -\frac{\beta(\theta)^2}{\alpha(\theta)} + \text{const}. \quad (8)$$

Let Θ denote the parameter domain of the ansatz family $\{|\phi(\theta)\rangle : \theta \in \Theta\}$. Assume that for each $\theta \in \Theta$, the inner minimum $\min_{t \in \mathbb{R}} \mathcal{L}(t, \theta)$ exists (which holds here because $\mathcal{L}(t, \theta)$ is a strictly convex quadratic in t with $\alpha(\theta) > 0$). Define $\tilde{\mathcal{L}}(\theta) := \min_{t \in \mathbb{R}} \mathcal{L}(t, \theta)$. Then the following identity is exact:

$$\min_{\theta \in \Theta} \min_{t \in \mathbb{R}} \mathcal{L}(t, \theta) = \min_{\theta \in \Theta} \tilde{\mathcal{L}}(\theta). \quad (9)$$

Proof. For every fixed $\theta \in \Theta$, by definition of $\tilde{\mathcal{L}}$ we have the pointwise equality $\min_{t \in \mathbb{R}} \mathcal{L}(t, \theta) = \tilde{\mathcal{L}}(\theta)$. Taking $\min_{\theta \in \Theta}$ of both sides preserves equality, giving (9). $\square$

Equation (9) formalizes why VQBR does *not* ignore the scale parameter: the scale is eliminated *exactly* via the closed-form minimizer $t^*(\theta)$ in (7), so optimizing only θ is equivalent to optimizing jointly over (t, θ) for the ansatz-restricted family $\{\mathbf{w} = t\phi(\theta)\}$. After training, a scaled estimate is recovered as

$$\hat{\mathbf{w}} = t^*(\theta^*)\phi(\theta^*), \quad \theta^* \in \arg \min_{\theta \in \Theta} \tilde{\mathcal{L}}(\theta). \quad (10)$$

If the ansatz family can represent the normalized MAP direction $\mathbf{w}^*/\|\mathbf{w}^*\|_2$, then $\phi(\theta^*)$ recovers the MAP direction; otherwise θ^* yields the minimum MAP objective value within the ansatz-restricted family.

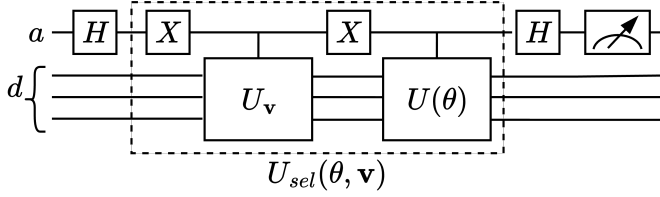


Fig. 1. Hadamard-test circuit used to estimate the overlap $\langle \mathbf{v} | \phi(\boldsymbol{\theta}) \rangle$. The controlled selection unitary $U_{\text{sel}}(\boldsymbol{\theta}, \mathbf{v})$ branches between $U_{\mathbf{v}}$ and $U(\boldsymbol{\theta})$ depending on the ancilla state. Measuring the ancilla after the final Hadamard yields $\Pr(z=0) = \frac{1+\langle \mathbf{v} | \phi(\boldsymbol{\theta}) \rangle}{2}$.

Define

$$\begin{aligned} a(\boldsymbol{\theta}) &:= \phi(\boldsymbol{\theta})^\top \mathbf{X}^\top \mathbf{X} \phi(\boldsymbol{\theta}), \\ c(\boldsymbol{\theta}) &:= \phi(\boldsymbol{\theta})^\top \mathbf{X}^\top \mathbf{y}, \\ d(\boldsymbol{\theta}) &:= \phi(\boldsymbol{\theta})^\top V \Sigma_0^{-1} \phi(\boldsymbol{\theta}), \\ e(\boldsymbol{\theta}) &:= \phi(\boldsymbol{\theta})^\top V \Sigma_0^{-1} \mathbf{m}_0. \end{aligned} \quad (11)$$

Therefore the

$$\tilde{\mathcal{L}}(\boldsymbol{\theta}) = -\frac{(c(\boldsymbol{\theta}) + e(\boldsymbol{\theta}))^2}{a(\boldsymbol{\theta}) + d(\boldsymbol{\theta})}. \quad (12)$$

C. Quantum operators and overlap estimation

1) *Quantum operators*: For each data row $\mathbf{x}_i^\top$, we build a unitary matrix $U_{\mathbf{x}_i} \in \mathbb{C}^{2^d \times 2^d}$ such that

$$U_{\mathbf{x}_i} |0\rangle^{\otimes d} = \left| \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2} \right\rangle. \quad (13)$$

For the prior cross term, we build $U_{\mathbf{c}} \in \mathbb{C}^{2^d \times 2^d}$ such that

$$U_{\mathbf{c}} |0\rangle^{\otimes d} = \left| \frac{\mathbf{c}}{\|\mathbf{c}\|_2} \right\rangle. \quad (14)$$

We choose an L -layer ansatz $U(\boldsymbol{\theta})$ on the same d -qubit register and set

$$|\phi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle^{\otimes d}. \quad (15)$$

In our implementation, we use real-amplitude encodings and a real-valued gate set (e.g., R_y and CNOT), so the overlaps estimated below are real.

2) *Overlap estimation*: Given a target unitary $U_{\mathbf{v}} \in \{U_{\mathbf{x}_i}, U_{\mathbf{c}}\}$, define

$$U_{\text{sel}}(\boldsymbol{\theta}, \mathbf{v}) = |0\rangle\langle 0| \otimes U_{\mathbf{v}} + |1\rangle\langle 1| \otimes U(\boldsymbol{\theta}). \quad (16)$$

We prepare $|0\rangle |0\rangle^{\otimes d}$, apply H to the ancilla, apply $U_{\text{sel}}(\boldsymbol{\theta}, \mathbf{v})$, apply H again to the ancilla, and measure the ancilla (Fig. 1). Let $z \in \{0, 1\}$ be the ancilla outcome and $|\mathbf{v}\rangle := U_{\mathbf{v}} |0\rangle^{\otimes d}$. Then

$$\begin{aligned} \Pr(z=0) &= \frac{1 + \langle \mathbf{v} | \phi(\boldsymbol{\theta}) \rangle}{2} \\ \Rightarrow \langle \mathbf{v} | \phi(\boldsymbol{\theta}) \rangle &= 2\Pr(z=0) - 1. \end{aligned} \quad (17)$$

D. Component estimation

1) *Computing $a(\boldsymbol{\theta})$ and $c(\boldsymbol{\theta})$* : For each data sample $\mathbf{x}_i$, using S shots of the overlap circuit with $U_{\mathbf{v}} = U_{\mathbf{x}_i}$, we estimate

$$\hat{s}_i(\boldsymbol{\theta}) = 2\frac{n_0^{(i)}}{S} - 1 \approx \left\langle \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}, \phi(\boldsymbol{\theta}) \right\rangle, \quad (18)$$

where $n_0^{(i)}$ is the number of times $z = 0$. Then, we estimate $a(\boldsymbol{\theta})$, and $c(\boldsymbol{\theta})$ as

$$\begin{aligned} \hat{a}(\boldsymbol{\theta}) &= \sum_{i=1}^N (\|\mathbf{x}_i\|_2 \hat{s}_i(\boldsymbol{\theta}))^2, \\ \hat{c}(\boldsymbol{\theta}) &= \sum_{i=1}^N y_i \|\mathbf{x}_i\|_2 \hat{s}_i(\boldsymbol{\theta}). \end{aligned} \quad (19)$$

2) *Estimating $d(\boldsymbol{\theta})$* : Let $M := V \Sigma_0^{-1}$. Since Σ_0^{-1} is symmetric positive definite, M is Hermitian and may therefore be regarded as a quantum observable on the d -qubit register. Thus,

$$d(\boldsymbol{\theta}) = \phi(\boldsymbol{\theta})^\top M \phi(\boldsymbol{\theta}) = \langle \phi(\boldsymbol{\theta}) | M | \phi(\boldsymbol{\theta}) \rangle. \quad (20)$$

To estimate this quantity, we decompose M into Pauli strings as

$$M = \sum_{\ell=1}^L \alpha_\ell P_\ell, \quad (21)$$

where each $P_\ell \in \{I, X, Y, Z\}^{\otimes d}$ is a Pauli string and $\alpha_\ell \in \mathbb{R}$. Then

$$d(\boldsymbol{\theta}) = \sum_{\ell=1}^L \alpha_\ell \langle \phi(\boldsymbol{\theta}) | P_\ell | \phi(\boldsymbol{\theta}) \rangle. \quad (22)$$

For each Pauli string P_ℓ , using S shots we estimate

$$\hat{\mu}_\ell(\boldsymbol{\theta}) \approx \langle \phi(\boldsymbol{\theta}) | P_\ell | \phi(\boldsymbol{\theta}) \rangle. \quad (23)$$

Then, we estimate $d(\boldsymbol{\theta})$ as

$$\hat{d}(\boldsymbol{\theta}) = \sum_{\ell=1}^L \alpha_\ell \hat{\mu}_\ell(\boldsymbol{\theta}). \quad (24)$$

3) *Estimating $e(\boldsymbol{\theta})$* : We estimate the prior cross term $e(\boldsymbol{\theta}) = \phi(\boldsymbol{\theta})^\top \mathbf{c}$ by running the overlap circuit with $U_{\mathbf{v}} = U_{\mathbf{c}}$. With $n_0^{(c)}$ the number of times $z = 0$ out of S shots,

$$\hat{h}(\boldsymbol{\theta}) = 2\frac{n_0^{(c)}}{S} - 1 \approx \left\langle \frac{\mathbf{c}}{\|\mathbf{c}\|_2}, \phi(\boldsymbol{\theta}) \right\rangle. \quad (25)$$

Then we estimate $e(\boldsymbol{\theta})$ as

$$\hat{e}(\boldsymbol{\theta}) = \|\mathbf{c}\|_2 \hat{h}(\boldsymbol{\theta}). \quad (26)$$

E. Variational quantum bayesian regression training

Need to repair

We minimize $\tilde{\mathcal{L}}(\boldsymbol{\theta})$ in (12) using a classical optimizer with quantum estimates in the loop. At each iteration we estimate $\hat{a}, \hat{c}, \hat{d}, \hat{e}$ using finite shots S , and evaluate $\tilde{\mathcal{L}}(\boldsymbol{\theta})$, then update $\boldsymbol{\theta}$. Algorithm 1 summarizes the hybrid training algorithm.

Algorithm 1 Variational Quantum Bayesian Regression

Require: Training data $(\mathbf{X}, \mathbf{y})$; diagonal prior $(\mathbf{m}_0, \Sigma_0)$; noise variance V ; iterations T ; batch size $|B|$; shots S ; ansatz $U(\theta)$; optimizer Opt .

Ensure: Trained state $|\phi(\theta^*)\rangle$ (and optional $\hat{\mathbf{w}}$)

- 1: Set $d = \lceil \log_2 D \rceil$ and zero-pad features to length 2^d if needed.
- 2: **for** $i = 1$ to N **do**
- 3: Build $U_{\mathbf{x}_i}$ such that $U_{\mathbf{x}_i} |0\rangle^{\otimes d} = |\bar{\mathbf{x}}_i\rangle$.
- 4: **end for**
- 5: $\mathbf{c} \leftarrow V\Sigma_0^{-1}\mathbf{m}_0$; build $U_{\mathbf{c}}$ such that $U_{\mathbf{c}} |0\rangle^{\otimes d} = |\bar{\mathbf{c}}\rangle$.
- 6: Initialize θ .
- 7: **for** $t = 1$ to T **do**
- 8: Sample minibatch $B \subseteq \{1, \dots, N\}$ with $|B|$ indices.
- 9: **for** each $i \in B$ **do**
- 10: Run overlap test with $U_{\mathbf{v}} = U_{\mathbf{x}_i}$ for S shots; set $\hat{s}_i(\theta) = 2n_0^{(i)}/S - 1$.
- 11: **end for**
- 12: Compute $\hat{a}(\theta), \hat{c}(\theta)$ via (19).
- 13: Run overlap test with $U_{\mathbf{v}} = U_{\mathbf{c}}$ for S shots; set $\hat{e}(\theta) = \|\mathbf{c}\|_2(2n_0^{(c)}/S - 1)$.
- 14: Measure $|\phi(\theta)\rangle$ in the computational basis for S shots; compute $\hat{d}(\theta)$ via (24).
- 15: Evaluate $\hat{\mathcal{L}}(\theta)$ via (12).
- 16: Update $\theta \leftarrow \text{Opt}(\hat{\mathcal{L}}, \theta)$.
- 17: **end for**
- 18: $\theta^* \leftarrow \theta$; prepare $|\phi(\theta^*)\rangle = U(\theta^*) |0\rangle^{\otimes d}$.
- 19: **return** $|\phi(\theta^*)\rangle$.

F. Complexity analysis

Need to repair

Each iteration requires $|B|$ overlap-estimation calls for data rows, one overlap-estimation call for the prior term, and one sampling routine for the diagonal prior quadratic, each with S shots. Thus the per-iteration measurement cost is

$$\mathcal{O}((|B| + 2)S), \quad (27)$$

up to constant factors.

The circuit depth is dominated by controlled state preparation for $U_{\mathbf{x}_i}$ and $U_{\mathbf{c}}$, and by the controlled ansatz $U(\theta)$ of depth L . In the explicit (compiled) state-preparation regime used in our NISQ experiments, $\text{Depth}(U_{\mathbf{x}_i})$ may scale with D in the worst case. Under oracle/qRAM-style data-access assumptions, state preparation is often modeled as $\text{polylog}(D)$, in which case the depth overhead from loading is correspondingly reduced. Finally, achieving additive precision $\pm\epsilon$ in overlap estimates via sampling typically requires $S = \mathcal{O}(1/\epsilon^2)$.

a) Comparison to classical methods.: For diagonal Σ_0 , the MAP optimizer solves the SPD system $\mathbf{A}\mathbf{w} = \mathbf{b}$, where $\mathbf{A} = \mathbf{X}^\top \mathbf{X} + V\Sigma_0^{-1}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y} + V\Sigma_0^{-1}\mathbf{m}_0$. A direct Cholesky-based solver requires forming $\mathbf{X}^\top \mathbf{X}$ ($\mathcal{O}(ND^2)$ time, $\mathcal{O}(D^2)$ memory) and factoring $\mathbf{A}$ ($\mathcal{O}(D^3)$ time, $\mathcal{O}(D^2)$ memory). Matrix-free iterative solvers (e.g., CG/LSQR) avoid explicit Gram construction; each iteration is dominated by

$\mathcal{O}(ND)$ -type matvecs (or $\mathcal{O}(\text{nnz}(\mathbf{X}))$ if sparse), with iteration counts governed by conditioning and preconditioning.

In contrast, VQBR avoids explicit Gram construction and replaces classical matvec-based updates with measurement-driven evaluation of the direction-only objective. Per outer iteration it performs $\mathcal{O}((|B| + 2)S)$ circuit executions, where the depth is dominated by controlled state preparation and the ansatz depth L . Consequently, VQBR trades classical arithmetic for shot budget and circuit depth, while producing a normalized direction state that supports inner-product queries without explicitly materializing all D coordinates.

b) Regimes with tangible benefit.: VQBR is most relevant when (i) D is large enough that explicitly storing/factorizing $\mathbf{A} \in \mathbb{R}^{D \times D}$ is prohibitive, (ii) the data/prior states are efficiently preparable (oracle/qRAM, structured/sparse amplitudes, or natively quantum-generated features), and (iii) downstream usage emphasizes prediction/inner-product access rather than full explicit recovery of $\mathbf{w}^*$. Conversely, for small/medium D , for tasks requiring explicit weights, or when hardware noise forces very large S for stable overlap estimates, classical direct or iterative solvers are typically preferable.

IV. NUMERICAL STUDIES

We evaluate VQBR on synthetic and real-world regression tasks with three goals: (i) verify correctness against the classical BLR/MAP closed form, (ii) compare optimization behavior under a matched iteration budget, and (iii) study robustness to data noise, finite-shot noise, ansatz depth, and encoding choices. Because VQBR outputs a *normalized* quantum state representing the MAP *direction*, our evaluation protocol includes a lightweight (evaluation-only) reconstruction step to obtain a comparable classical weight vector for standard regression metrics.

A. Experimental settings

1) Datasets:

a) Synthetic data: We generate $\mathbf{X} \in \mathbb{R}^{N \times D}$ with i.i.d. standard normal entries and sample targets via

$$\mathbf{y} = \mathbf{X}\mathbf{w}_{\text{true}} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (28)$$

We sweep both feature dimension and sample size:

- **Dimension sweep:** $D \in \{8, 16, 32, 64, 128\}$, with $d = \lceil \log_2 D \rceil$ and zero-padding to length 2^d when needed.
- **Sample-size sweep:** $N \in \{2D, 5D, 10D\}$.

For each (N, D) pair, we use an 80%/20% train/test split; all hyperparameters are selected using training data only.

b) Real-world data: We additionally evaluate on three regression benchmarks spanning small to larger feature dimensions: (i) *Energy Efficiency* [18] ($N = 768, D = 8$), (ii) *Naval Propulsion Plants* [19] ($N = 11,934, D = 16$), and (iii) *YearPredictionMSD* [20] ($N = 515,345, D = 90$). For each dataset, features are standardized using training-set statistics only (zero mean, unit variance), then split into 80%/20% train/test.

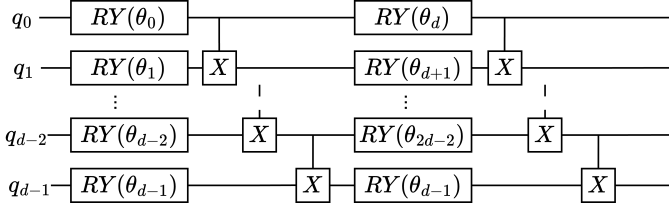


Fig. 2. Parameterized ansatz $U(\theta)$ implemented using `EfficientSU2` with single-qubit R_y rotations and linear entanglement. The circuit consists of two repetition layers ($\text{reps} = 2$), where each layer applies parameterized R_y rotations on all qubits followed by nearest-neighbor CNOT entangling gates in a linear topology.

2) *Prior and noise specification:* We use a diagonal Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_0, \Sigma_0)$ with $\Sigma_0 = \text{diag}(\sigma_1^2, \dots, \sigma_D^2)$ as in standard BLR [1], [2]. To model feature-wise shrinkage, we set $\Sigma_0^{-1} = \text{diag}(\lambda_1, \dots, \lambda_D)$ using training statistics only:

$$\lambda_j = \lambda_0 \frac{1}{\widehat{\text{Var}}(X_{:,j}) + \delta}, \quad j = 1, \dots, D, \quad (29)$$

with $\lambda_0 > 0$ chosen using training data only.

In all experiments, Σ_0^{-1} (and hence $\mathbf{c} = V\Sigma_0^{-1}\mathbf{m}_0$) is computed exactly from training statistics using classical preprocessing; we do not include this preprocessing cost in the VQBR training loop budget.

a) *Noise variance.:* For synthetic experiments, $V = \sigma^2$ is known by construction and provided to all methods. For real-world datasets, V is unknown and is selected from training data only. Concretely, we carve out a validation subset from the training split, sweep a small grid of candidate values for (V, λ) (isotropic) or (V, λ_0) (heteroscedastic), and select the pair minimizing validation MSE. We then refit on the full training split using the selected hyperparameters and report test performance.

3) *Baselines and evaluation budgets:* We compare VQBR to classical solvers that optimize the same BLR/MAP objective (Problem 1). Unless stated otherwise, all methods use the same training split and prior $(\mathbf{m}_0, \Sigma_0)$.

a) (i) *Closed-form BLR/MAP:* We compute the MAP solution

$$\mathbf{w}^* = \left(\mathbf{X}^\top \mathbf{X} + V\Sigma_0^{-1} \right)^{-1} \left(\mathbf{X}^\top \mathbf{y} + V\Sigma_0^{-1}\mathbf{m}_0 \right), \quad (30)$$

using an SPD solver (Cholesky on $\mathbf{A} = \mathbf{X}^\top \mathbf{X} + V\Sigma_0^{-1}$). This serves as the accuracy reference.

b) (iii) *Proposed VQBR:* We train VQBR by minimizing the direction-only objective $\widehat{\mathcal{L}}(\theta)$ (Sec. III-B) using full-batch COBYLA. In particular, we run COBYLA using the entire training set at each iteration (i.e., $B = \{1, \dots, N_{\text{train}}\}$). Unless stated otherwise, $U(\theta)$ is `EfficientSU2` (Fig. 2) with $\text{reps} = 2$ (linear entanglement), and we use $S = 4096$ shots per circuit call. COBYLA is gradient-free, avoiding additional measurements for gradient estimation, and is often stable under finite-shot noise [21], [22].

4) *Evaluation protocol:* VQBR outputs a trained circuit $U(\theta^*)$ preparing a *normalized* direction state $|\phi(\theta^*)\rangle$ whose amplitudes approximate $\mathbf{w}^*/\|\mathbf{w}^*\|_2$. To compute standard regression metrics (MSE, ℓ_2 error) that require an explicit weight vector, we form a classical proxy $\widehat{\mathbf{w}}$ from the trained quantum state using an evaluation-only reconstruction. In particular, we use an exact statevector simulator, we directly read the amplitude vector $\phi(\theta^*) \in \mathbb{R}^{2^d}$ and truncate/unpad to the first D entries.

Because the learned state encodes a *direction*, we rescale it to obtain a fair, comparable estimator in $\mathbb{R}^D$. In this work, we use the closed-form magnitude as a consistent reference scaling:

$$\widehat{\mathbf{w}} = \|\mathbf{w}^*\|_2 \cdot \phi_D(\theta^*), \quad (31)$$

where $\phi_D(\theta^*)$ denotes the D -dimensional unpadded amplitude vector. This choice isolates the quality of the *learned direction* (the primary output of VQBR) while enabling standard regression metrics. Importantly, VQBR is a *hybrid quantum-classical* algorithm whose primary deliverable is the trained circuit $U(\theta^*)$; we treat the above reconstruction/rescaling strictly as an *evaluation protocol*, and do not optimize or claim efficiency for this post-processing step.

5) *Metrics:* We report four metrics, computed on both train and test splits when applicable:

- **Cosine similarity (direction alignment):**

$$\cos(\widehat{\mathbf{w}}, \mathbf{w}^*) = \frac{\widehat{\mathbf{w}}^\top \mathbf{w}^*}{\|\widehat{\mathbf{w}}\|_2 \|\mathbf{w}^*\|_2}. \quad (32)$$

- **Train/Test MSE:**

$$\begin{aligned} \text{MSE}_{\text{train}} &= \frac{1}{N_{\text{train}}} \|\mathbf{X}_{\text{train}} \widehat{\mathbf{w}} - \mathbf{y}_{\text{train}}\|_2^2, \\ \text{MSE}_{\text{test}} &= \frac{1}{N_{\text{test}}} \|\mathbf{X}_{\text{test}} \widehat{\mathbf{w}} - \mathbf{y}_{\text{test}}\|_2^2. \end{aligned} \quad (33)$$

- **Relative ℓ_2 parameter error:**

$$\text{RelErr} = \frac{\|\widehat{\mathbf{w}} - \mathbf{w}^*\|_2}{\|\mathbf{w}^*\|_2}. \quad (34)$$

Cosine similarity is the primary indicator of whether the trained circuit captures the MAP *direction*, while MSE and ℓ_2 errors quantify downstream predictive fidelity of the reconstructed classical proxy $\widehat{\mathbf{w}}$ under the evaluation protocol above.

B. Results

1) *Results on synthetic data:* Table I reports the synthetic-data scaling results for VQBR trained with COBYLA, aggregated over 20 random seeds (mean $\pm$ std). We quantify agreement with the classical closed-form MAP solution using cosine similarity (directional alignment) and relative ℓ_2 distance (magnitude-sensitive discrepancy), and we additionally report test MSE for both VQBR and the closed-form baseline. The qubit count is $\lceil \log_2 D \rceil$, corresponding to the amplitude-encoding dimension used by the proposed formulation.

Two consistent trends emerge. First, in low-dimensional regimes VQBR attains strong alignment with the closed-form MAP solution. For example, at $D = 8$ and $N = 80$,

TABLE I
SYNTHETIC SCALING STUDY OF VQBR. COSINE SIMILARITY AND RELATIVE ℓ_2 DISTANCE ARE COMPUTED WITH RESPECT TO THE CLASSICAL CLOSED-FORM MAP SOLUTION.

(N, D)	Qubits	Cosine $\uparrow$	Rel. $\ell_2 \downarrow$	Test MSE (VQBR) $\downarrow$	Test MSE (Closed-form) $\downarrow$
$D = 8$ (3 qubits)					
(16, 8)	3	0.787 ± 0.192	0.594 ± 0.272	3.59 ± 3.34	0.660 ± 2.360
(40, 8)	3	0.925 ± 0.065	0.339 ± 0.187	1.04 ± 0.97	0.015 ± 0.022
(80, 8)	3	0.963 ± 0.045	0.225 ± 0.151	0.56 ± 0.57	0.012 ± 0.005
$D = 16$ (4 qubits)					
(32, 16)	4	0.636 ± 0.172	0.827 ± 0.207	9.22 ± 8.68	0.056 ± 0.069
(80, 16)	4	0.725 ± 0.132	0.718 ± 0.186	8.53 ± 6.38	0.016 ± 0.006
(160, 16)	4	0.806 ± 0.092	0.604 ± 0.153	7.01 ± 4.15	0.012 ± 0.002
$D = 32$ (5 qubits)					
(64, 32)	5	0.540 ± 0.144	0.948 ± 0.151	31.29 ± 17.97	0.028 ± 0.012
(160, 32)	5	0.628 ± 0.084	0.857 ± 0.093	24.01 ± 7.58	0.014 ± 0.003
(320, 32)	5	0.639 ± 0.085	0.844 ± 0.100	25.65 ± 9.19	0.012 ± 0.002
$D = 64$ (6 qubits)					
(128, 64)	6	0.389 ± 0.072	1.104 ± 0.065	69.41 ± 21.87	0.027 ± 0.010
(320, 64)	6	0.468 ± 0.083	1.028 ± 0.082	71.57 ± 18.89	0.013 ± 0.002
(640, 64)	6	0.474 ± 0.085	1.022 ± 0.085	68.63 ± 17.81	0.012 ± 0.001
$D = 128$ (7 qubits)					
(256, 128)	7	0.307 ± 0.083	1.176 ± 0.070	160.51 ± 43.76	0.028 ± 0.007
(640, 128)	7	0.342 ± 0.080	1.145 ± 0.069	174.96 ± 34.26	0.013 ± 0.002
(1280, 128)	7	0.377 ± 0.052	1.116 ± 0.047	153.13 ± 24.59	0.012 ± 0.001

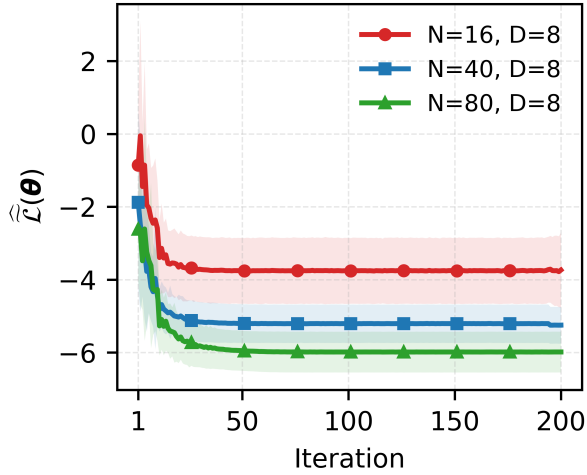


Fig. 3. Training convergence of the direction-only objective $\hat{\mathcal{L}}(\theta)$ for $D = 8$ (3 qubits) under COBYLA. Curves show the mean over 20 random seeds; shaded regions denote ± 1 standard deviation. Increasing N yields a lower (better) objective value and faster stabilization.

VQBR reaches cosine similarity 0.963 ± 0.045 with relative ℓ_2 distance 0.225 ± 0.151 , and achieves test MSE 0.56 ± 0.57 (Table I). Moreover, for fixed D , increasing the sample size N improves both alignment metrics and predictive performance, indicating that the variational training benefits from additional data. This behavior is also reflected in Fig. 3, where larger N leads to a lower stabilized objective value and reduced variability across seeds.

Second, as the feature dimension D increases (and thus the number of qubits grows), the approximation quality degrades under a fixed ansatz depth and optimization budget. For $D \in \{32, 64, 128\}$, cosine similarity decreases while relative ℓ_2 distance often exceeds 1, and the VQBR test MSE becomes substantially larger than the closed-form baseline (Table I). This trend is consistent with the increased optimization difficulty of wider amplitude-encoded variational circuits on NISQ-scale settings, motivating the ansatz/optimizer refinements examined in subsequent experiments.

2) *Result on real-world dataset:* Table II summarizes performance on both synthetic and real-world benchmarks. We report training and testing MSE, together with direction-level agreement to the closed-form MAP solution via cosine similarity and parameter-error metrics (absolute and relative ℓ_2 error). For VQBR, metrics are computed using the evaluation-only reconstruction described in Sec. IV-A4.

To characterize optimization behavior, Fig. 4 and Fig. 5 plot the convergence of the direction-only objective $\hat{\mathcal{L}}(\theta)$ and the direction alignment (cosine similarity) versus COBYLA iterations. Unless stated otherwise, VQBR uses full-batch COBYLA with $S = 4096$ shots per circuit call and the EfficientSU2 ansatz with $\text{reps} = 2$.

V. CONCLUSION

We presented VQBR, a variational quantum training framework for Bayesian regression that learns a circuit encoding the MAP solution direction while incorporating prior knowledge through measurement-defined objective estimators. Future work includes posterior-uncertainty extraction beyond

TABLE II

PERFORMANCE SUMMARY ON SYNTHETIC AND REAL-WORLD BENCHMARKS (PLACEHOLDERS). FOR EACH DATASET, WE COMPARE THE CLOSED-FORM BLR/MAP REFERENCE TO THE PROPOSED VQBR. COSINE SIMILARITY IS MEASURED AGAINST THE CLOSED-FORM DIRECTION; MSEs ARE COMPUTED USING THE RECONSTRUCTED $\hat{\mathbf{w}}$ (SEC. IV-A4).

Dataset	Method	Train MSE	Test MSE	Cosine $\uparrow$	Rel. ℓ_2 Err $\downarrow$
Synthetic ($N = \cdot$, $D = \cdot$)	Closed-form MAP	[—]	[—]	1.000	0.000
	VQBR	[—]	[—]	[—]	[—]
Energy ($N = 768$, $D = 8$)	Closed-form MAP	[—]	[—]	1.000	0.000
	VQBR	[—]	[—]	[—]	[—]
Naval ($N = 11,934$, $D = 16$)	Closed-form MAP	[—]	[—]	1.000	0.000
	VQBR	[—]	[—]	[—]	[—]
YearPredictionMSD ($N = 515,345$, $D = 90$)	Closed-form MAP	[—]	[—]	1.000	0.000
	VQBR (COBYLA, full-batch)	[—]	[—]	[—]	[—]

figures/convergence_synth_objective.pdf

figures/convergence_synth_cosine.pdf

Fig. 4. Synthetic convergence (placeholders). Left: direction-only objective $\hat{\mathcal{L}}(\theta)$ versus COBYLA iterations. Right: cosine similarity between the learned direction $\phi(\theta)$ and the closed-form MAP direction $\mathbf{w}^*/\|\mathbf{w}^*\|_2$.

MAP (e.g., approximate predictive variance) and structured-prior implementations with tighter operator-access guarantees.

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figures/convergence_real_cosine.pdf

Fig. 5. Real-world convergence (placeholders). Left: $\widehat{\mathcal{L}}(\theta)$ versus COBYLA iterations for each benchmark. Right: cosine similarity versus iterations. Curves should be averaged over seeds (if applicable) with error bars or shaded standard deviation.

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